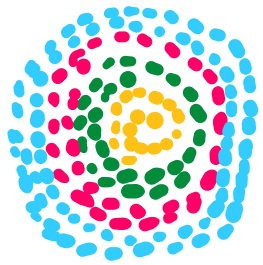
Neural Networks-1

* Limitations of ML Algorithms :-

Is this data linear →

Ans - No.



Data
↓
Labelled
↓
Linear X
↓
Clean X

	Multi-Class	Non-Linear	Problem
Logistic Regression	Cannot Work	Cannot Work	Cannot create complex Decision Boundary
KNN	Can Work	Can Handle	• Inference time is high if data inc • Other classes may appear similar due to non-linear data
Decision Tree	Can Work	Can handle	Cannot work well with text, speech, image, sparse data (high dimensional data)
SVM	Cannot Work	Can Handle using Kernel func	Cannot work with big datasets

Therefore taking Logistic Regression (Linear Regression in case of

Therefore taking Logistic Regression (Linear Regression in case of continuous O/P) and making it more powerful

Powerful → ① Automate complex feature creation

② Adaptive for Multi-class Classification

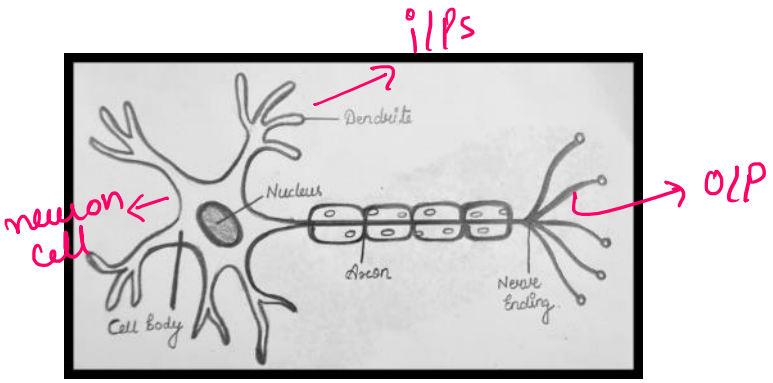
These powerful models are nothing but known as

"NEURAL NETWORKS"

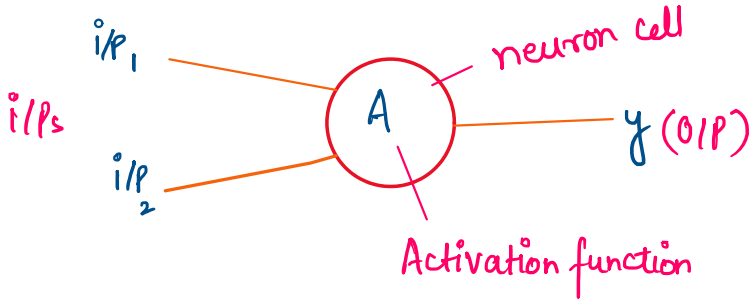
Neural Network :-

It is a type of a ML model that is inspired by a human brain.

★ Neuron:-



Biological Neuron



Artificial Neuron

Biological Neuron → Some i/Ps are of high thickness

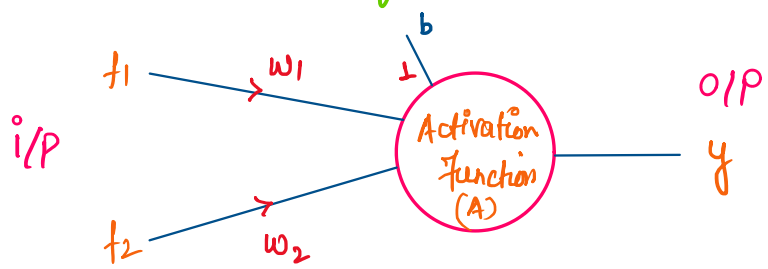
Biological Neuron → some i/ps are of high thickness
" " " thin

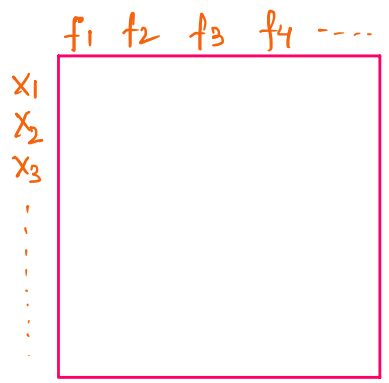
To apply the same logic in ANN, weights are applied.

Important i/p → ↑weight

Less " " → ↓ weight

How does an artificial Neuron looks like?





These i/ps are the features

* Applications of NN in our day to day life :-

1. Responsive ads to show relevant info.
2. Data Compression [data of millions of users due to social media]
3. Image enhancement
4. Text Autocompletion in emails
5. chatbots

* History of NN :-

First simplest model → 1957 (Rosenblatt)] → loosely inspired from biological neuron

Human Brain has multiple neurons

So, the researchers created an artifical neural network

As complexity of NN increased, it didn't perform well.

Then comes a break through with the concept of "Backpropa-gation". But this hype of AI died again in 1990s

↳ Lack of computational power

↳ Data

That was known as the period of AI winter

Then in 2012, Imagenet started a competition

Stanford Univ.

↓

Deep NN created using GPUs and sufficient data

Where can we use NN?

1. Predicting continuous data
2. Classification
3. Unsupervised learning [Forecasting, Recomm., Clustering]
4. Dealing with image, text, speech data.

④ Dealing with image, text, speech data.

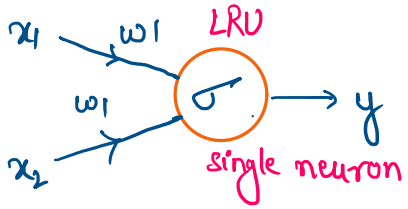
★ Artificial Neuron using single logistic model :-

↓

Binary classification → 0 or 1

↓

prob. (z) > 0.5 → 1 else 0



→ y → prob → 0 or 1

x_1 & x_2 are features, σ = sigmoid, y = O/P

What about multiclass classification?

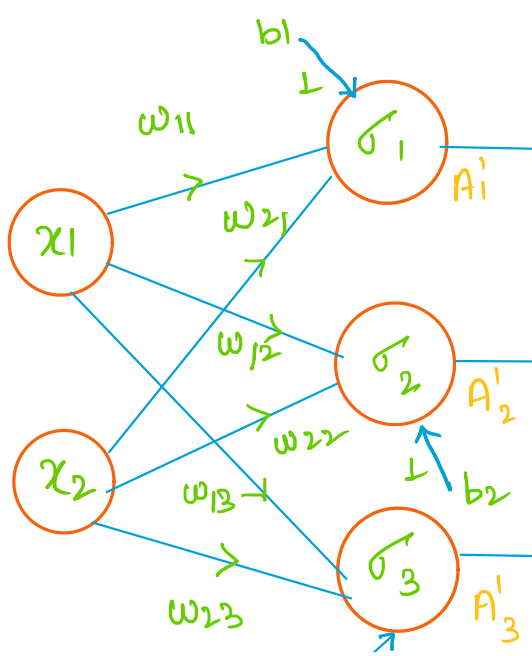
↓

Multiple Logistic Regression Units (LRUs)

★ Neural Network for Multiclass Classification :-

class = 3 [M, N, O], i/Ps = 2 (features)

For this we combine 3 LRUs to create a NN



M

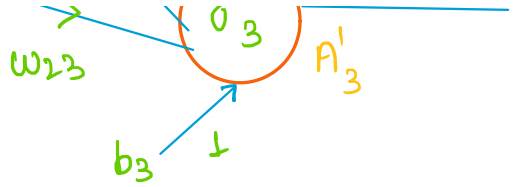
$y_1 = P_1$ (>0.5 → 1 else 0)

N

$y_2 = P_2$ (>0.5 → 1 else 0)

O

$y_3 = P_3$ (>0.5 → 1 else 0)



$\hat{y}_3 = P_3$ ($> 0.5 \rightarrow 1$ else 0)

∵ there are 3 classes to classify.

∴ we use 3 neurons in o/p layer

Notations :-

① Inputs :- x_{i1}, x_{i2}

∵ we have two features, so we take two i/ps

But we have m datapoints, so, there will be an i/p data matrix

$$\text{Datapoint Matrix} = \begin{bmatrix} x_{11} & x_{12} \\ x_{21} & x_{22} \\ x_{31} & x_{32} \\ \vdots & \vdots \\ x_{m1} & x_{m2} \end{bmatrix}_{m \times 2}$$

② Neurons :-

No. of neurons depends on the layer.

For i/p layer → neuron = no. of features

For o/p layer → neuron = no. of classes

Each neuron (except the ones in the i/p layer) consists of an activation function

General representation of neuron → A'_1, A'_2, A'_3 .... etc.

$A' \rightarrow 1$ here represents the layer

$A^{l} \rightarrow l$ here represents the layer

③ Weights :- w_{ij}

∵ the computation is happening together in NN,

∴ we take a weight matrix rather than a weight vector

ex-

ML	NN
say → f_1 f_2 f_3	say → f_1 f_2 f_3
weight → vector $\begin{bmatrix} w_1 \\ w_2 \\ w_3 \end{bmatrix}$	we assign multiple wts. to one feature
	weight → matrix $\begin{bmatrix} w_{11} & w_{12} & w_{13} \\ w_{21} & w_{22} & w_{23} \end{bmatrix}$
	$W^T = \begin{bmatrix} w_{11} & w_{21} \\ w_{12} & w_{22} \\ w_{13} & w_{23} \end{bmatrix}$

$w_{ij} \Rightarrow$ $i \rightarrow$ source neuron

$j \rightarrow$ destination neuron

④ Bias :- b_i

Each neuron will have a bias term

$$b = [b_1 \quad b_2 \quad b_3]_{1\times 3}$$

⑤ Z-Value :- It is weighted sum of i/ps with their respective weights

(5) Z-Value :- It is weighted sum of i/Ps with their respective weights

$$Z_1 = w_{11}x_1 + w_{21}x_2 + b_1$$
$$Z_2 = w_{12}x_1 + w_{22}x_2 + b_2$$
$$Z_3 = w_{13}x_1 + w_{23}x_2 + b_3$$

(6) Output :- Output of each neuron will be the result of an activation function

Let sigmoid be the activation →

$$A_1' = \sigma(Z_1) \quad ; \quad A_2' = \sigma(Z_2) \quad ; \quad A_3' = \sigma(Z_3)$$

We are using NN to achieve non-linearity

Neuron = Linear + Activation

In NN computation is done by passing the value of i/Ps & their respective weights from 1st layer to 2nd layer & so on...

Moving from i/P to o/P in neural network is called as forward propagation

In forward propagation, we calculate the predicted value & the loss.

Similarly, moving from o/P to i/P in NN is called Back Propagation.

During back propagation, weights are updated.

★ Can we use any other function instead of sigmoid?

Ans. Yes

★ Issue with sigmoid :-

When applied on 3 different neurons, it will result with probabilities like -

$$P_1 = 0.8 \quad , \quad P_2 = 0.7 \quad , \quad P_3 = 0.4$$

The total prob. will be > 1

This means that the NN is predicting prob > 0.5 (threshold) for more than 1 classes.

So, for multiclass classification, we need a function that distributes the probabilities properly accross all classes.

∴ That function is → "Softmax classifier"